

Ghilmo Choi

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Education

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| Korea Aerospace University , Republic of Korea
B.S. in Mechanical Engineering | Mar 2021 – Feb 2027
(expected) |
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- Overall GPA: **4.08 / 4.5** (Major GPA: **4.18 / 4.5**)
 - Main Coursework: Introduction to Machine Learning, Automatic Control, Mechanical Vibrations, Data Structures and Algorithms, Probabilistic Robotics (in progress)

Research Interests

- On-device vision-language models for natural-language robotic mission execution
- Perception, Prediction, and Decision Making for Autonomous Driving
- Motion planning and path tracking for real-vehicle autonomous driving

Research / Project Experience

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| ASOL (Autonomous Systems and Optimization Lab)
- <i>Undergraduate Researcher</i> | Korea Aerospace Univ.
June 2025 – Present |
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- On-device VLM-based robotic mission execution**
Developed a natural-language mission execution framework that converts user commands into executable robot behaviors on a Traxxas TRX-4 & UGV
 - Hallucination-aware and SLAM-grounded task planning**
Designed an Actor–Guard–Critic planning loop and integrated RTAB-Map, object detection, structured world states, and ROS2/Nav2 control for real-robot language-to-motion execution
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| KAUVOY Autonomous Driving Club
- <i>Localization Part</i> | Korea Aerospace Univ.
Nov 2024 – Nov 2025 |
|---|--|
- EKF-based localization**
Implemented Dual EKF localization on an ERP42 autonomous vehicle by fusing IMU, wheel encoder, and RTK-GPS measurements
 - Autonomous parking and path tracking**
Developed and tested a real-vehicle parking pipeline covering cone perception, parking-space decision, Hybrid A* path planning, and Pure Pursuit-based path tracking
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| - <i>Team Leader</i> | Nov 2025 – present |
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- Finite-State-Machine-based decision-making module for autonomous driving scenarios**
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| Course Project, Introduction to Machine Learning
- End-to-End Machine Learning Classification and Regression Pipeline | Korea Aerospace Univ.
Aug 2025 – Nov 2025 |
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- Built and validated classification and regression pipelines using preprocessing, dimensionality reduction, ensemble methods, and FCNNs; achieved **0.915 macro F1** on a 4-class imbalanced classification task

Honors & Activities

- Finalist, **2025 College Mobility Innovation Competition**
- Excellence Award, **Drone X AI Marketing Challenge** (2026)

Publications

- **Co-author**, "*On-Device Vision Language Model Based Natural Language Mission Execution Framework*," submitted to the International Conference on Control, Automation and Information Sciences (**ICCAIS**), 2025
- **Co-author**, "*ROS `robot_localization` Dual EKF: Verification of Localization Accuracy and Application to Autonomous Driving Path Tracking*," submitted to the Korean Society of Automotive Engineers (**KSAE**), 2025

Technical Skills

- Python, C/C++, Linux, ROS, Docker, Git